Payout Mechanisms as Operators on Attention Distributions: Identities, Invariances, and Tail Action

Tonify Research

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How to read this paper. A streaming platform pools subscriber money and splits it among artists; the three ways to split are *pro-rata* (one pool, paid by global play share), *user-centric* (each wallet split only across its owner's own plays), and *direct* (no pool: fans pay artists directly). This paper treats all three as operators mapping a play matrix to artist incomes and answers four questions: who gains from a pro-rata → user-centric switch (Theorem 1); whether any pool-rule reform can touch the signed-vs-independent gap (Lemma 2); whether the direct economy reduces top-end concentration (it does not — it relocates inequality to a mass of exact zeros); and whether direct income can ever stochastically dominate the pool (it cannot — Theorem 4). Every claim below is machine-verified in the companion code and survived two adversarial red-team passes. The italicized paragraphs ("Why this matters" / "What it says") form a complete non-technical route: a reader may skip every formula and proof (set in small type) and still carry away the results.

1 Setup and the generative model of sim4

Artists $i \in N$, listeners $u \in U$ ($|U| = U$), integer play matrix $p = (p_{iu}) \in \mathbb{Z}_+^{N\times U}$ over one period. Row/column sums: $P_i = \sum_u p_{iu}$ (the artist's plays), $P_u = \sum_i p_{iu}$ (the listener's intensity), $T = \sum_u P_u$, mean intensity $\bar{P} = T/U$. Equal wallets $W > 0$; pool $R = UW$.

Assumption 1 (A1, load-bearing)**.** *Every paying subscriber listens: $P_u \geq 1$. Silent payers are excluded from* both *pools; otherwise budget balance below fails by exactly $W \cdot U_{\mathrm{sil}}$ and all ratios acquire the factor $U_{\mathrm{act}}/U_{\mathrm{tot}}$.*

Mechanisms. Pro-rata $\mathrm{PR}_i = UW\,P_i/T$; user-centric $\mathrm{UC}_i = W\sum_u p_{iu}/P_u$; direct $D_i = \tau \sum_{f=1}^{S_i}\sum_{t=1}^{k} G_{ft}$ with $S_i \mid A_i \sim \mathrm{Bin}(A_i, \sigma)$, audience A_i (the number of listeners with $p_{iu} > 0$), superfan share σ, i.i.d. gifts $G \geq 0$, $\mathbb{E}G = \bar{g}$, all moments finite; artist share τ. A label contract acts multiplicatively after the rule: $\rho \cdot F_i(p)$.

Generative model (sim4). Artist popularity: piecewise $s_i \sim$ (lognormal body / log-bridge / Pareto(α=1.4) tail), calibrated to anchors T_1: 87% of artists below 10^3 streams/yr, T_2: 2.6% above 225,734, T_3: top 0.28% holding 40–55% of streams (gate band; the seed-42 matrix realizes 40.3%, sim1's larger world 44.5%); target audiences $\ell_i = s_i/\mathrm{PL}$, PL = 21.21 plays per pair (LFM-1b). Playlist sizes $K_u \sim \mathrm{LogNormal}(\ln 12, 1.0)$ (assumed; an MVP-measured dial). Artist selection:

exact weighted sampling without replacement, $\mathbb{P}(i \in S_u) \propto \ell_i$ (Gumbel top-K_u), i.e. preferential attachment. Pair intensity, three named regimes on one graph:

(a) ergodic: integer p_{iu}, $P_u \equiv 2048$; (b) $p_{iu} = \lceil \mathrm{LN}(\ln 5.16,\, 1.676) \rceil$ rescaled to s_i; (c) $p_{iu} \propto \mathrm{LN}\cdot(A_i/\bar{A}_g)^\gamma$.

Regime (a) has $P_u \equiv 2048$ exactly (a power of two: every p_{iu}/P_u is a dyadic rational, so the control identity is exact in floating point) and is the control (Corollary (a)); γ dials the coupling between pair intensity and audience size — its role becomes central in the tail-action analysis.

2 The ratio identity

Why this matters. *Debates about user-centric run on intuition ("it helps niche artists"). The identity below replaces intuition with a formula: an artist's user-centric-to-pro-rata income ratio is a specific average over that artist's own audience, so the winners of a reform can be computed before the reform.*

Theorem 1 (Identity)**.** *For any artist i with $P_i > 0$, under equal wallets,*

$$\frac{\mathrm{UC}_i}{\mathrm{PR}_i} = \mathbb{E}_{w^i}\left[\frac{\bar{P}}{P_u}\right], \qquad w^i_u = \frac{p_{iu}}{P_i}.$$

Proof. $\mathrm{UC}_i/\mathrm{PR}_i = \big(W \sum_u p_{iu}/P_u\big)\big/\big(UWP_i/T\big) = (T/U)\sum_u (p_{iu}/P_i)(1/P_u) = \bar{P}\,\mathbb{E}_{w^i}[1/P_u]$. □

With $m_i = \mathbb{E}_{w^i}[P_u]$ (arithmetic) and $H_i = 1/\mathbb{E}_{w^i}[1/P_u]$ (harmonic mean of listener intensity under play weights),

$$\frac{\mathrm{UC}_i}{\mathrm{PR}_i} = \frac{\bar{P}}{H_i} = \underbrace{\frac{\bar{P}}{m_i}}_{\text{level}} \cdot \underbrace{\frac{m_i}{H_i}}_{\text{dispersion } \geq 1},$$

the second factor ≥ 1 by AM–HM, with equality iff P_u is constant on the support of w^i.

What it says. *The ratio splits into level × dispersion: a "light" audience (below-average listeners) helps through the first factor, and a* heterogeneous *audience always helps through the second. So the folk claim "user-centric helps the niche" is wrong in an instructive way: what helps is not nicheness but lightness and heterogeneity of one's listeners.*

Corollary (a) (Coincidence; the naive converse is false)**.** $P_u \equiv \bar{P}$ *implies* UC = PR *componentwise. The converse fails: with one artist and two listeners, $p = (10, 20)$, intensities differ yet* $\mathrm{UC}_1 = \mathrm{PR}_1 = 2W$*. Exact criterion:* $\mathrm{UC}_i = \mathrm{PR}_i$ *for all i iff the vector $c_u = 1/P_u - U/T$ lies in* $\ker p$.

Corollary (b) (Zero sum)**.** $\sum_i \mathrm{UC}_i = \sum_i \mathrm{PR}_i = UW$*: user-centric is a pure redistribution of the same pool.*

Corollary (c) (Crossover)**.** $\mathrm{UC}_i > \mathrm{PR}_i$ *iff $H_i < \bar{P}$. On type populations this reduces to the condition of Alaei, Makhdoumi, Malekian and Pekeč (*Mgmt. Sci. *2022, Props. 2, 8): artist j weakly prefers pro-rata iff $\sum_i q_i \pi_{ij}(\lambda_i - \bar{\lambda}) \geq 0$.*

Corollary (d) (Dispersion premium)**.** *Fix $m_i = \bar{P}$. Then $\mathrm{UC}_i/\mathrm{PR}_i = m_i/H_i \geq 1$, with equality iff audience intensity is degenerate: at the same mean level,* within-audience dispersion strictly raises *user-centric income.*

3 Pass-through invariance

Why this matters. *The central policy dispute: is the artist squeezed by the pool rule or by the label contract? If a signed artist keeps seven cents on the dollar, can any reform of the splitting rule close that gap? The lemma answers: never in income terms — and sometimes in minimum-viable-audience terms.*

Lemma 2 (Two-part invariance)**.** (2a) *For any pool operator F* applied at the artist level*, with the multiplicative contract strictly downstream and on the domain $F_i(p) > 0$: $\rho F_i/F_i = \rho$ — the signed/independent* income *gap is invariant to the rule (a nonlinear F applied to a licensor's* catalogue *breaks this).* (2b) *The* minimum-viable-audience *gap satisfies* $\mathrm{MVA}_s/\mathrm{MVA}_\iota = 1/\rho$ *iff* $F^{-1}(V/\rho) = F^{-1}(V)/\rho$*; linearity of F-income in audience is sufficient but* not *necessary (a strictly increasing log-periodic F is nonlinear yet yields $1/\rho$ for all V). A Deezer-style $\times 2$ boost above a threshold yields $1/(2\rho)$ instead (counterexample in companion code) while the income ratio stays ρ pointwise.*

What it says. *A rule reform cannot move the contractual income gap by one cent; it can move the "minimum audience to make a living" gap, but only by breaking the scaling criterion above. This formalizes the empirical regularity that label shares did not move across rule changes (Pedersen 2020). Breaking channels: endogenous $\rho(F)$; additive components; payout thresholds (accumulation delays $\times 1/\rho$ in time); behavioral endogeneity of p; component/market-heterogeneous $\rho_{\text{eff}} = \sum_m \rho_m w_{im}(F)$; payee-level aggregation of a nonlinear rule.*

4 Tail action and the atom at zero

Why this matters. *Streaming's chief indictment is concentration: the top percent takes nearly everything. Would a direct fan-to-artist economy spread income more evenly? This section answers for the top and the bottom of the distribution separately — and the answer disappoints every mechanism.*

Let $\mathbb{P}(A > x) = x^{-\alpha}L(x)$. PR is linear: index preserved. UC: $\mathrm{UC}_i = WP_i\,\mathbb{E}_{w^i}[1/P_u]$ with bounded multiplier; under tail-conditional concentration of the multiplier the index is preserved, and heavier tails are impossible for any dependence. Direct: binomial thinning plus a light-tailed compound sum preserve α (Jessen–Mikosch 2006, Lem. 3.7; Faÿ et al. 2006; Denisov–Foss–Korshunov 2010; Breiman 1965 enters only through the multiplicative contract layer), while $\mathbb{P}(D_i = 0 \,|\, A_i) \geq (1-\sigma)^{A_i} \geq e^{-1}(1-\sigma)$ for $A_i \leq 1/\sigma$: an *atom at zero* on small audiences. With zero mass q, $G_{\text{direct}} = q + (1-q)G^{+}$: the mechanism relocates inequality from the intensive to the extensive margin; no mechanism softens the upper tail.

What it says. *No mechanism touches how much the top takes; what changes is the* shape of the bottom*. Pool rules pay many tiny positive amounts; the direct economy converts most of those into exact zeros — at the calibrated superfan share, at least a third of small-audience artists earn literally nothing. Inequality is relocated, not reduced.*

5 Dominance blockade

Why this matters. *Suppose the direct economy pays four times the pool on average. Should a small artist leave the pool? Not necessarily: the average hides a large chance of zero. This section compares the two incomes by their whole distributions, in the standard language of stochastic dominance.*

Fix A with $(1-\sigma)^A > 0$ and pooled income $X_A = cA > 0$ (deterministic given A); $Y_A = D_A$ as above.

Theorem 4 (Full blockade)**.** *(i) Y_A never first-order dominates X_A; (ii) Y_A never second-order dominates X_A, for any parameters; (iii) $X_A \succeq_2 Y_A$ iff $\mathbb{E}Y_A \leq X_A$, i.e. iff $k \leq d^* = \mathrm{PL{\cdot}rate}/(\sigma\bar{g}\tau)$; (iv) for $k > d^*$ the pair is incomparable in both orders.*

Proof. (i) For $x \in (0, X_A)$: $\mathbb{P}(X_A > x) = 1$ while $\mathbb{P}(Y_A > x) \leq 1 - (1-\sigma)^A < 1$. (ii) SSD against degenerate X_A requires $\int_0^t F_Y \leq \max(0, t - X_A)$; at $t = X_A$ the left side is $\geq (1-\sigma)^A X_A > 0$. (iii) ($\Leftarrow$) $\int_0^t F_Y = t - \mathbb{E}\min(Y,t) \geq t - \mathbb{E}Y \geq t - X_A$ and ≥ 0. ($\Rightarrow$) $t \to \infty$ gives $\mathbb{E}Y \leq X_A$. (iv) From (i)–(iii). $\square$

Remark. The correct comparison is mean versus risk: for $k > d^*$ direct pays an expectation premium priced by small-audience lottery risk; $\mathbb{P}(Y_A < X_A) \leq \inf_{t>0} e^{tX_A}\big(1 - \sigma + \sigma\varphi(t)^k\big)^A$ with $\varphi(t) = \mathbb{E}e^{-t\tau G} < 1$ — exponential decay in A. In the companion sim1 calibration at A=30, k=4: mean advantage $\times 4$, yet $\mathbb{P}(Y < X) \approx 0.60$ — and on the full 2M-sample run $\mathbb{P}(Y < X) - \mathbb{P}(Y = 0) = 0$ to machine precision: losing to the pool is exactly the zero outcome.

What it says. *The zero atom blocks stochastic dominance in both directions and in both orders, for all parameter values: the direct economy can never be unconditionally better than the pool, no matter how generous the fans. The honest frame is mean versus risk — direct pays an expectation premium priced by a lottery on small audiences, and only raising the superfan conversion σ (not payment frequency k) attacks the lottery itself.*